

Walkshed and Accessibility Analysis: A Foundation for Sustainable and Inclusive Urban Mobility in Bhopal City

Satyam Kumar

1. Introduction & Context

Walking is the primary human transport mode, available to all, except those who are walking impaired (Vasconcellos, 2005).

This report presents a comprehensive walkshed and accessibility analysis for Bhopal City, aiming to provide data-driven insights for the development of sustainable and inclusive urban mobility strategies.

1.1 Background

Rapid urbanization globally, including in Indian cities like Bhopal, necessitates a shift towards more sustainable and equitable urban development. Traditional planning often prioritizes motorized transport, leading to issues such as traffic congestion, air pollution, increased carbon emissions, and reduced quality of life. Recognizing these challenges, there is a growing emphasis on promoting active transportation modes, particularly walking, as a cornerstone of healthy, vibrant, and environmentally responsible cities.

1.2 Aim and Objectives of the Study

The overall aim of this study is to develop a detailed understanding and visual representation of the pedestrian-accessible areas within Bhopal, originating from specific points of interest or encompassing defined time-based walking catchments.

The specific objectives are:

- To quantify the extent of pedestrian-accessible areas in Bhopal within 5, 10, and 15-minute walking durations.
- To analyse the distribution of different land use categories within these pedestrian walksheds.
- To evaluate the quality of pedestrian accessibility across various road types in Bhopal.

- To spatially map and quantify the distribution of high, medium, low, and no-access areas across different land use categories.
- To assess the pedestrian accessibility around existing bus stops and metro stations.
- To provide data-driven recommendations for improving walkability and accessibility to foster sustainable and inclusive urban mobility in Bhopal.

1.3 Study Area: Bhopal City

Bhopal, the capital city of Madhya Pradesh, India, is a rapidly developing urban centre facing typical challenges associated with growth, including increasing traffic and demand for infrastructure. Its unique geography, including two large lakes (Upper Lake and Lower Lake), influences its urban form and connectivity. Understanding pedestrian mobility in Bhopal is crucial for its sustainable and inclusive growth, ensuring that its development benefits all residents.

2. Methodology

This analysis employed Geographic Information Systems (GIS) techniques to conduct watershed and accessibility assessments, integrating various spatial datasets of Bhopal City.

2.1 Data Sources

The primary data sources utilized for this study included:

- Road network data.
- Land use maps of Bhopal City.
- Geospatial data for existing bus stops.
- Geospatial data for proposed metro station locations.

2.2 Walkshed Analysis

Walkshed analysis was performed using network analysis tool in QGIS. This involved:

- Isochrone Generation: Isochrones (areas reachable within a specified travel time) were generated from key origin points like all existing bus stops and all proposed metro stations.
- Time Increments: Walksheds were calculated for 5-minute, 10-minute, and 15-minute walking durations, representing typical pedestrian travel times for short to medium trips.

- **Area Calculation:** The total accessible area (in km²) and its percentage of the total study area were calculated for each time increment.
- **Land Use Intersect:** The generated walkshed polygons were intersected with the land use map to determine the area and percentage of each land use category falling within the respective walking times.

2.3 Accessibility Analysis

The assessment of accessibility levels (High, Medium, Low, No Access) for road types and land use areas was based on a combination of factors related to pedestrian infrastructure quality and connectivity.

- For the "Accessibility Map", Space Syntax analysis was employed to calculate Mean Angular Depth that is a measure of integration in a spatial network, indicating how easily a space is reached from all other spaces in the system. Lower values typically indicate higher integration and accessibility.

3. Walksheds around Public Transport Hubs

Understanding the pedestrian catchment areas around existing public transport nodes is crucial for enhancing first-and-last-mile connectivity and promoting multi-modal travel.

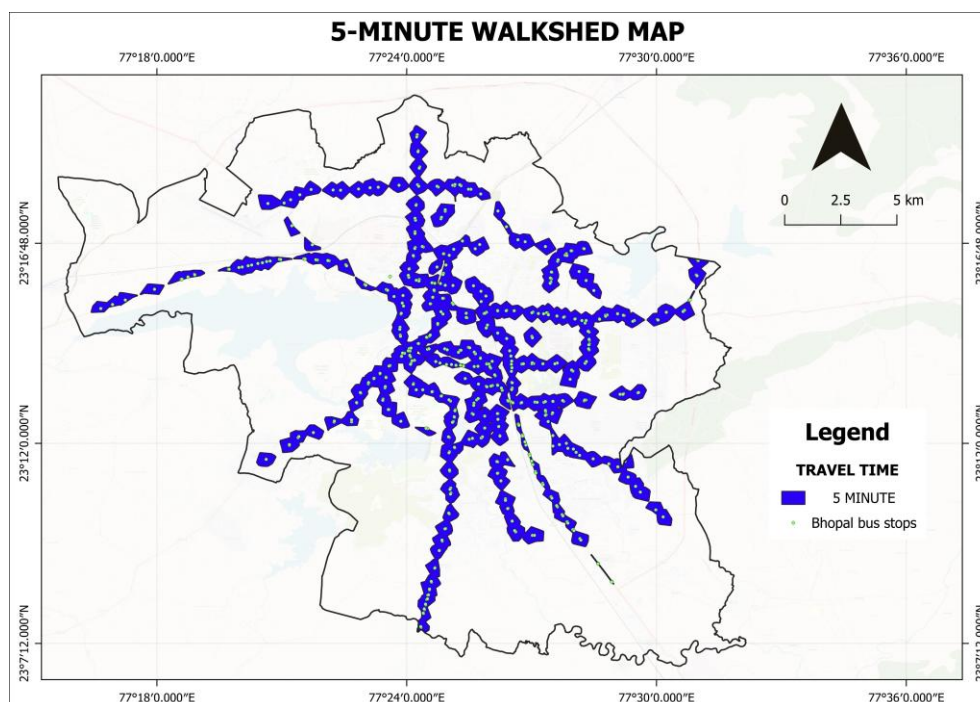


Figure 3.1: 5-Minute Walkshed Map around Bhopal Bus Stops

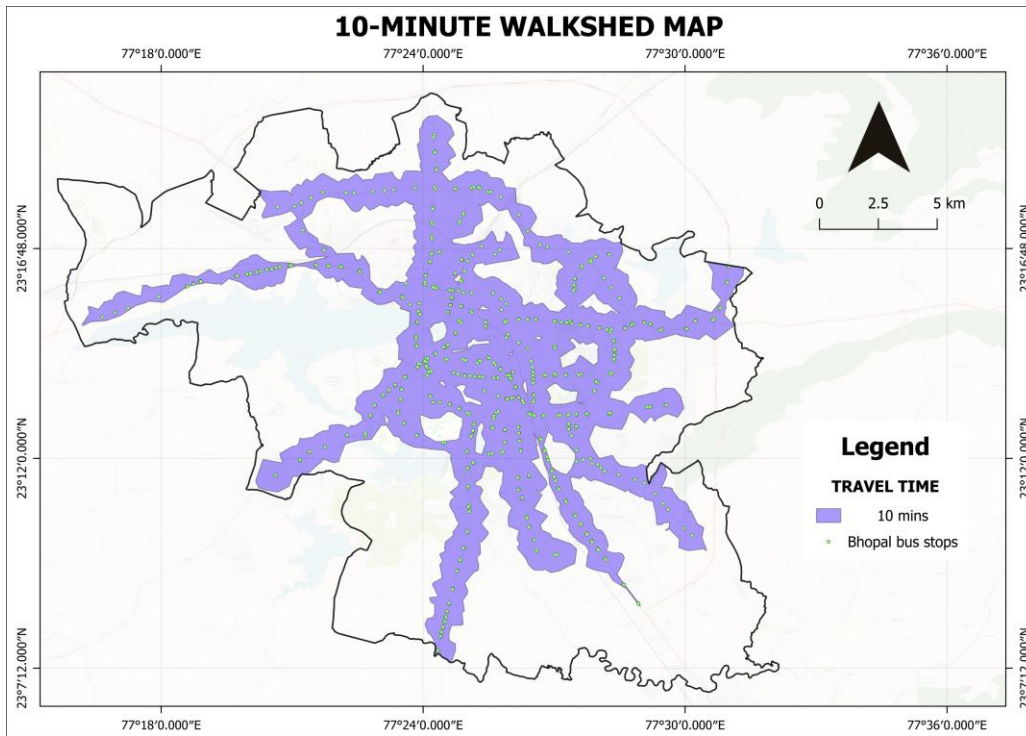


Figure 3.2: 10-Minute Walkshed Map around Bhopal Bus Stops

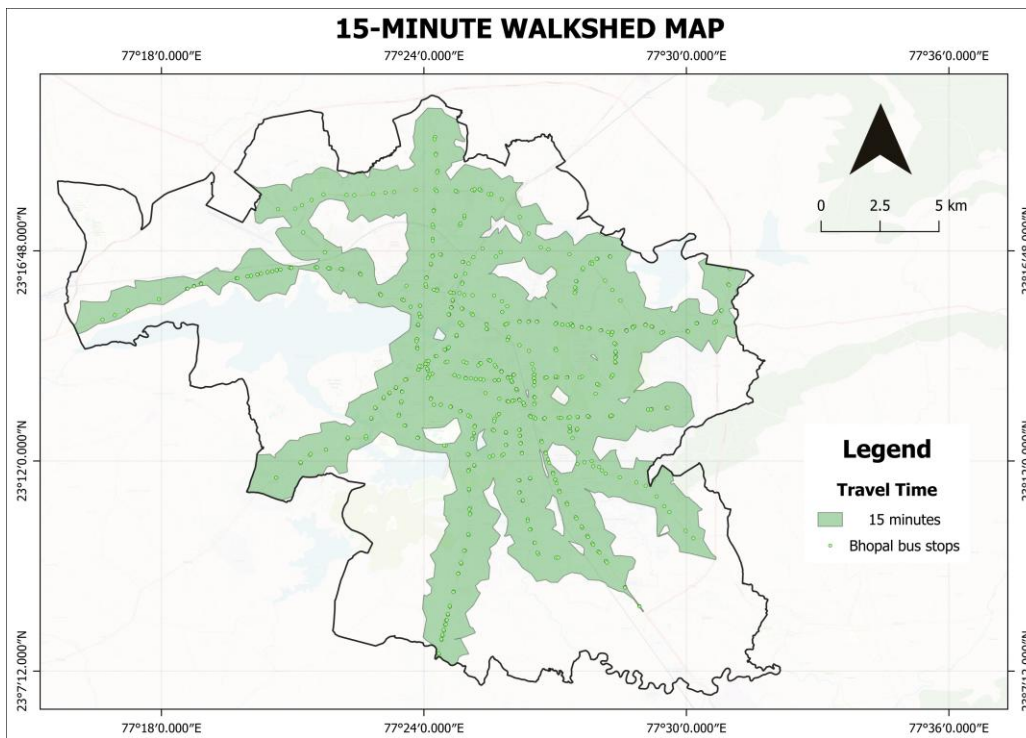


Figure 3.3: 15-Minute Walkshed Map around Bhopal Bus Stops

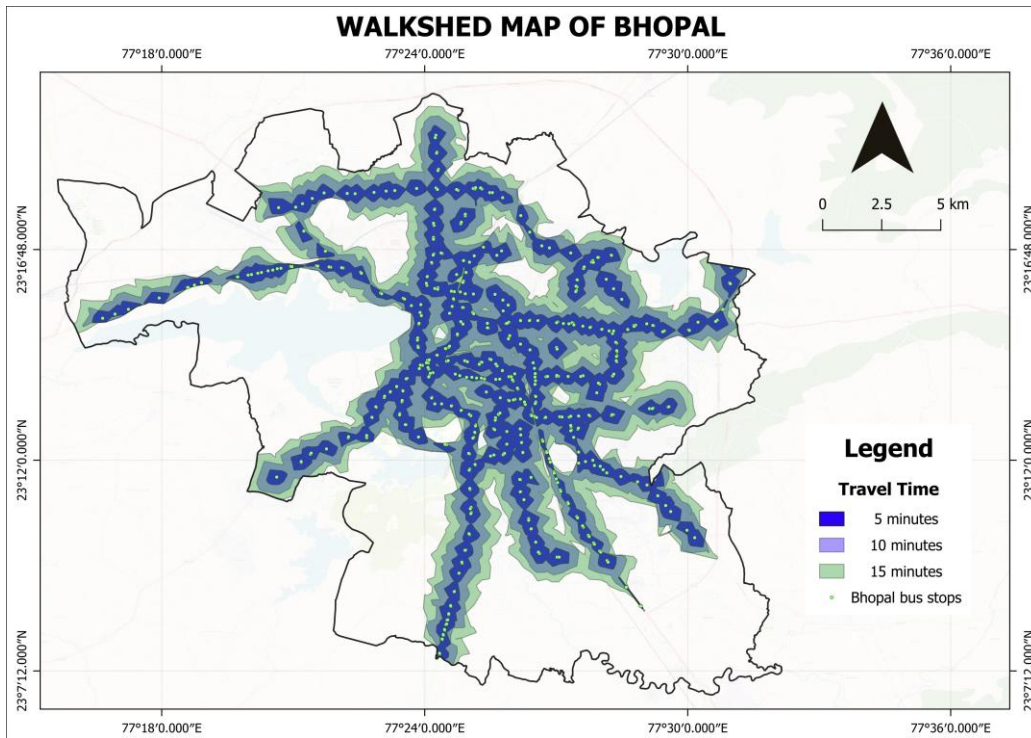


Figure 3.4: Overall Walkshed Map of Bhopal around Bus Stops (Combined)

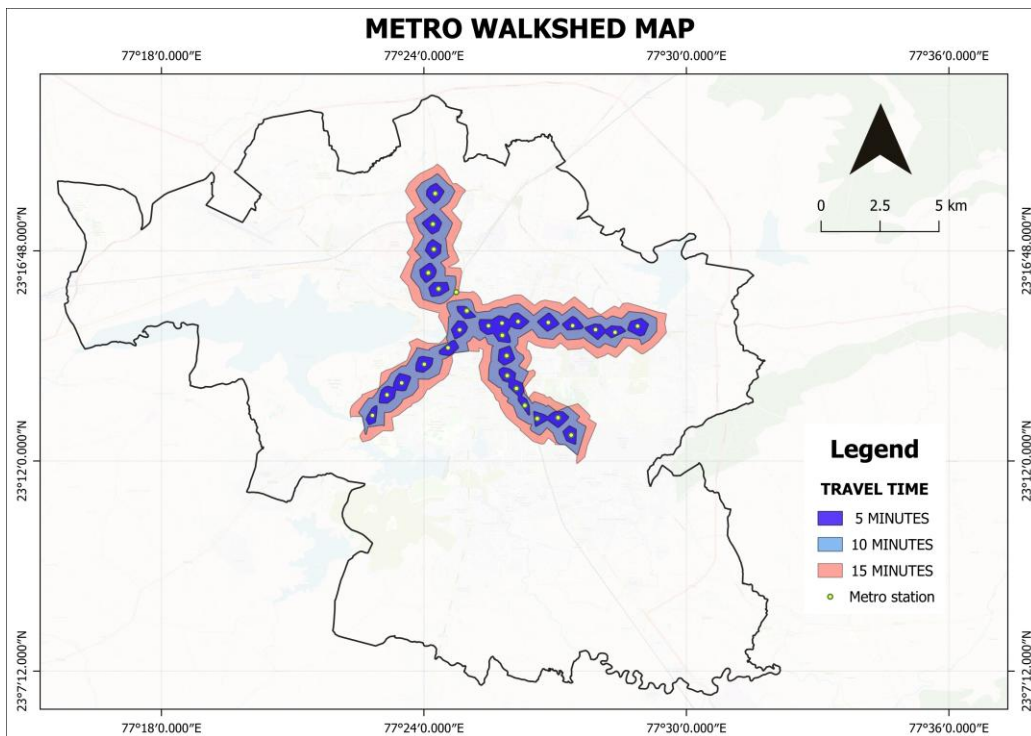


Figure 3.5: 5, 10, and 15-Minute Walkshed Map around Bhopal Metro Stations

4. Walkshed Analysis Findings

This section presents the findings related to the spatial extent of areas accessible by walking within Bhopal, providing a foundational understanding of pedestrian mobility potential.

4.1 Overall Pedestrian Coverage

The analysis of pedestrian accessibility coverage across Bhopal City reveals the proportion of the total urban area that is reachable within standard walking durations.

Pedestrian Accessibility Coverage in Bhopal by Walking Time

Walking Duration	Travel Mode	Accessible Area (km ²)	Percentage of Total Area
5 minutes	Foot-walking	74.364	16.59%
10 minutes	Foot-walking	155.658	34.72%
15 minutes	Foot-walking	206.974	46.17%

Table 4.1: Overall Pedestrian Accessibility Coverage in Bhopal by Walking Time

Table 4.1 demonstrates that a significant portion of Bhopal's total area is accessible by foot within typical walking times. Approximately 16.59% (74.364 km²) of the city is within a 5-minute walk, expanding to 34.72% (155.658 km²) within 10 minutes, and nearly half the city, 46.17% (206.974 km²), within a 15-minute walk. This indicates a substantial existing potential for pedestrian-based mobility for shorter trips across the city, highlighting the importance of fostering and improving this mode of transport for sustainable urban development.

4.2 Accessible Land Use Distribution by Walking Time

This analysis integrates walkshed data with land use information, providing insights into the accessibility of different urban functions on foot.

Percentage of Land Use Areas by Walking Time

Land Use Category	Total Area (km²)	5-min Walk (%)	10-min Walk (%)	15-min Walk (%)
Agriculture	140.418	3.52	10.11	18.85
Barren/Wasteland	34.361	9.41	27.57	42.59
Commercial	25.920	41.46	69.03	79.52
Green Cover	36.111	7.03	20.36	30.99
Industrial	7.468	17.99	43.75	67.47
Public/Semi-Public	38.893	24.15	46.23	56.34
Public Utilities	1.780	41.07	60.00	72.19
Residential	117.962	27.61	55.64	68.47
Slum	5.658	27.38	61.26	76.60
Water Bodies	38.903	2.58	5.45	8.09

Table 4.2: Percentage of Different Land Use Areas Accessible by Walking Time in Bhopal

Table 4.2 provides critical insights into which types of land uses are predominantly within walkable distances in Bhopal. For instance, a substantial 69.03% of commercial areas are accessible within a 10-minute walk, indicating a high potential for pedestrian activity in these economic zones. Similarly, public utilities show high accessibility (60.00% within 10 min). However, residential areas, while having over half (55.64%) accessible within 10 minutes, show room for improvement to ensure daily needs are within very short walking distances. The relatively low accessibility for agriculture and water bodies is expected, as these are not primary destinations for everyday walking trips. Crucially, the data for 'Slum' areas (61.26% within 10 min) suggests a reasonable level of access, but further investigation into the quality of this access is warranted to ensure true inclusivity.

5. Accessibility and Connectivity Analysis

Space Syntax analysis, specifically Mean Angular Depth, provides a network-based understanding of spatial integration and connectivity, indicating how easily a space is reached from all other spaces in the system.

5.1 Accessibility Map of Bhopal City

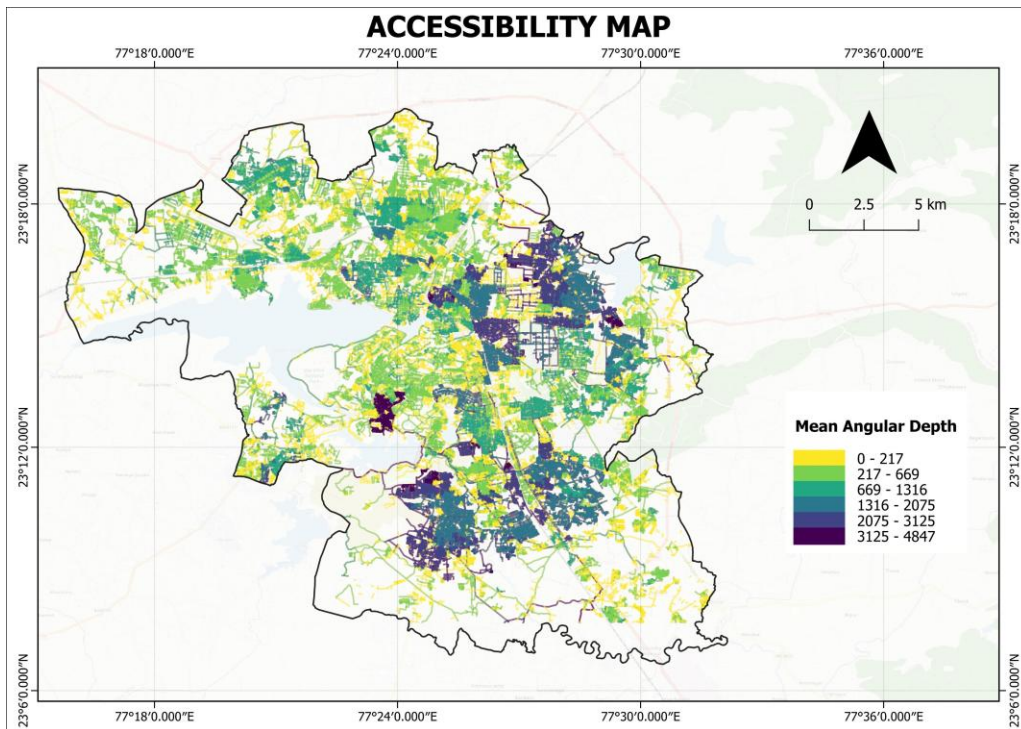


Figure 5.1: Accessibility Map of Bhopal based on Mean Angular Depth

Figure 5.1, representing the Mean Angular Depth of Bhopal's street network, visually reinforces the city's spatial integration and pedestrian connectivity. Areas depicted in [e.g., yellow, indicating lower Mean Angular Depth values] are highly integrated and easily reachable from most parts of the network, signifying areas of high pedestrian potential and likely higher foot traffic. These often correspond to central business districts or well-connected residential neighbourhoods. Conversely, areas depicted in [e.g., dark blue/purple, indicating higher Mean Angular Depth values] represent less integrated spaces, which might be harder to navigate on foot, leading to lower pedestrian activity. This map is a powerful tool for urban planners to identify areas where network connectivity improvements (e.g., creating new pedestrian links, improving intersections, removing physical barriers) are most needed to enhance overall accessibility and foster more fluid pedestrian movement throughout the city.

5.2 Accessibility Analysis Findings: Accessibility by Road Type

The quality of pedestrian infrastructure varies significantly across different road types, directly impacting the safety, comfort, and desirability of walking.

Accessibility Levels by Road Type in Bhopal

Road Type	High Access (%)	Medium Access (%)	Low Access (%)	No Access (%)
Footway	87.81	8.46	0.00	3.73
Living Street	68.63	13.73	1.96	15.69
Path	68.75	18.75	3.12	9.38
Pedestrian	50.00	0.00	0.00	50.00
Residential	33.88	47.92	17.08	1.12
Service	79.58	9.66	2.12	8.64
Tertiary	27.62	48.33	23.84	0.22
Unclassified	72.87	15.17	9.89	2.07

Table 5.2: Pedestrian Accessibility Levels by Road Type in Bhopal

Table 5.2 reveals significant disparities in pedestrian access quality across different road types in Bhopal. Dedicated 'footways' demonstrate excellent high access (87.81%), indicating that where dedicated pedestrian paths exist, they are generally well-designed. 'Service' roads also show high access (79.58%), suggesting good pedestrian conditions in these localized service areas. However, 'Residential' roads, which constitute a large part of the daily walking environment for citizens, show only 33.88% high access, with a large proportion falling into 'Medium' (47.92%) and 'Low' (17.08%) access categories. Similarly, 'Tertiary' roads, important for connecting neighbourhoods, also have substantial 'Medium' (48.33%) and 'Low' (23.84%) access. This highlights a widespread deficit in basic, high-quality pedestrian infrastructure where people live and move daily. The 50% 'No Access' for 'pedestrian' classified roads is particularly concerning, suggesting either incomplete pedestrian zones or significant barriers. This data is critical for prioritizing infrastructure investments to improve the everyday walking experience and safety.

5.3 Land Use Area Distribution by Accessibility Level

This analysis quantifies the physical area of different land use categories that fall into various accessibility tiers, providing a spatial understanding of where high-quality pedestrian environments are present or lacking.

Land Use Area Distribution by Accessibility Level (in km²)

Land Use Category	HIGH	LOW	MEDIUM	NO ACCESS	Grand Total
Agriculture	116.28	13.109	8.331	–	137.72
Barren/Wasteland	12.694	9.458	11.939	–	34.091
Commercial	14.494	3.252	2.447	4.901	25.094
Green Cover	18.643	4.498	10.238	0.287	33.666
Industrial	1.683	–	5.362	–	7.045
Public Utilities	–	0.107	0.173	–	0.28
Public/Semi-Public	20.18	3.099	15.548	–	38.827
Residential	53.513	29.744	33.252	0.153	116.662
Slum	1.816	0.42	3.514	–	5.75
Water Bodies	–	33.981	1.512	–	35.493
Grand Total	239.303	97.668	92.316	5.341	434.628

Table 5.3: Distribution of Land Use Areas by Pedestrian Accessibility Level in Bhopal (in km²)

Table 5.3 provides a crucial spatial quantification of accessibility challenges and successes across Bhopal's land uses. While a substantial 239.303 km² of the total area has 'High' pedestrian access, a significant 97.668 km² falls into 'Low' access and 92.316 km² into 'Medium' access categories.

- **Residential Areas:** A key finding is that out of 116.662 km² of residential land, 29.744 km² has 'Low' access and 33.252 km² has 'Medium' access. This means over 63 km² of residential areas suffer from suboptimal pedestrian environments, directly impacting the daily mobility and quality of life for a large segment of Bhopal's population.
- **Commercial Areas:** While 14.494 km² of commercial land has 'High' access, the presence of 4.901 km² with 'No Access' is concerning. This indicates severe pedestrian bottlenecks that can fragment the urban network, limit customer access, and hinder economic activity in these specific commercial zones.
- **Public/Semi-Public:** While a good portion (20.18 km²) has 'High' access, 15.548 km² falls into 'Medium' access, suggesting that reaching some public services might still be inconvenient for many.

- **Slum Areas:** With 1.816 km² of 'High' access and 3.514 km² of 'Medium' access, it indicates some pedestrian connectivity, but improvements are needed to ensure equitable and high-quality access to essential services for these communities.

6. Role in Planning Sustainable and Inclusive Urban Mobility for Bhopal

The comprehensive walkshed and accessibility analysis presented in this report provides critical insights that directly inform the planning of sustainable and inclusive urban mobility in Bhopal.

6.1 Promoting Sustainable Mobility

The analysis quantifies the existing potential for walking as a primary mode of transport in Bhopal. By identifying that nearly half of the city's area is within a 15-minute walk, and that a significant portion of commercial and public utility areas are highly accessible on foot, the report underscores the feasibility of reducing reliance on private vehicles for short to medium trips. However, the presence of extensive 'Low' and 'Medium' access areas in residential zones and on key road types indicates where current infrastructure deficiencies force residents into motorized transport. Improving these areas will directly contribute to reduced traffic congestion, lower fuel consumption, and decreased carbon emissions, aligning with Bhopal's sustainability goals. The Mean Angular Depth map further guides efforts to enhance network efficiency, making walking a more attractive and viable option.

6.2 Enhancing Inclusive Mobility

A key contribution of this analysis is its ability to highlight disparities in access, which is fundamental to inclusive urban planning. The comparison between overall accessibility and potentially lower accessibility in specific land uses reveals areas where certain populations may face greater challenges in reaching essential services and opportunities. The significant proportion of 'Low' and 'Medium' access in residential and slum areas points to specific equity concerns, where vulnerable populations might lack safe and convenient pedestrian access. By identifying these gaps, the report provides a basis for targeted interventions that ensure all residents, regardless of their location or socioeconomic status, have equitable access to the city's resources, fostering social cohesion and reducing urban inequalities.

6.3 Integration with Public Transport

The walkshed analysis around bus stops and metro stations is vital for optimizing public transport usage. By improving pedestrian links to these transit hubs, Bhopal can enhance first-and-last-mile connectivity, making public transport more attractive and convenient.

7. Recommendations

Based on the findings of this walkshed and accessibility analysis, the following recommendations are put forth to foster sustainable and inclusive urban mobility in Bhopal:

7.1 Prioritize Pedestrian Infrastructure Upgrades

- **Target Residential and Tertiary Roads:** Focus immediate investment on improving pedestrian infrastructure (e.g., continuous, wide, well-maintained sidewalks; safe crossings) in the significant areas of 'Residential' and 'Tertiary' roads identified with 'Low' and 'Medium' access. This will directly enhance daily walkability for a large segment of the population.
- **Address "No Access" Bottlenecks:** Conduct detailed site-specific studies for the 'No Access' areas, particularly within 'Commercial' land uses, to identify and implement solutions such as new pedestrian bridges, underpasses, or traffic calming measures to create vital connections.
- **Improve Network Connectivity:** Utilize the Mean Angular Depth map to identify less integrated areas and plan for new pedestrian pathways or improved intersections that enhance overall network connectivity and make walking more intuitive.

7.2 Promote Mixed-Use Development

- **Strategic Zoning:** Encourage and incentivize mixed-use zoning, especially in areas currently lacking essential services within short walking distances of residential zones. This will reduce the need for motorized trips for daily errands and foster more self-sufficient neighbourhoods.
- **Transit-Oriented Development (TOD):** Ensure that new developments around metro stations are designed with high walkability and mixed-use principles, integrating residential, commercial, and public spaces to maximize the benefits of public transport investment.

7.3 Enhance Public Transport Accessibility

- **First-and-Last-Mile Connectivity:** Implement programs to improve pedestrian access to all existing bus stops and future metro stations. This includes ensuring safe, direct, and well-lit pathways, clear signage, and accessible crossings.
- **Shelter and Amenities:** Provide adequate shelters, seating, and information at public transport stops to enhance the pedestrian experience.

7.4 Ensure Equitable Access

- **Targeted Interventions:** Prioritize infrastructure improvements in underserved communities, including slum areas, to bridge existing accessibility gaps and ensure equitable access to opportunities for all residents.

- Universal Design Principles: Integrate universal design principles into all pedestrian infrastructure projects to ensure accessibility for people of all ages and abilities, including those with mobility impairments.

8. Conclusion

This walkshed and accessibility analysis for Bhopal City underscores the critical role of pedestrian mobility in achieving sustainable and inclusive urban development. The findings highlight both the existing potential for walking as a primary mode of transport and significant areas where infrastructure deficiencies and spatial barriers impede pedestrian movement. By quantifying accessible areas, evaluating infrastructure quality, and identifying specific zones of concern, this report provides a robust, data-driven foundation for strategic urban planning. Implementing the recommended interventions, focusing on targeted infrastructure upgrades, promoting mixed-use development, enhancing public transport integration, and ensuring equitable access, will enable Bhopal to foster a more walkable, liveable, environmentally friendly, and socially just urban environment for all its citizens.

References

- Software/GIS platform used is QGIS.
- Source of road network data from OpenStreetMap and Bhopal Municipal Corporation.
- Land use data collected through primary survey.